Reinforcement learning is a recent arrival in macroeconomics. Most of the algorithms surveyed in this chapter were published in the past few years, and the literature is small enough that few results have been independently replicated. The arrival is nevertheless of some interest, because RL admits two distinct interpretations in a macro context. As a solver, RL is a computational tool that finds rational-expectations or Nash equilibria for models whose state space breaks dynamic programming. As a behavioural model, RL describes a boundedly-rational agent who learns a policy from realised utility under no a priori knowledge of the economy. The two interpretations share the same training code; the distinction is in what the analyst uses the output to claim.

This chapter is organised around four lenses on what RL does for macroeconomics. Section 0.1 fixes notation and defines the four problem classes the chapter uses. Section 0.2 and Section 0.3 treat RL as a global solver, first for representative-agent macro models that classical dynamic programming already handles, and then for heterogeneous-agent models where DP breaks down. The simulation in Section 0.4 concretises the comparison on a textbook RBC. Section 0.5 takes the continuum limit to mean field games with common noise, in two variants by game structure: symmetric mean-field and hierarchical Stackelberg. Section 0.7 reframes RL as a model of the household itself, using learning dynamics to reproduce empirical anomalies that rational expectations cannot match jointly. Section 0.8 surveys RL for policy design under two methodologically distinct settings, multi-agent ABM in which the planner and the citizens are all RL learners co-adapting, and single-planner RL in which only the planner uses RL and the rest of the macro model is fixed. Section 0.10 closes with a short synthesis.

0.1 Setup and notation

We collect the definitions here so that subsequent sections can map a paper's notation to a single chapter-wide convention without restating the machinery.

0.1.1 The agent's MDP

A Markov decision process (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \beta)$ collecting state space, action space, transition rule, reward, and discount. Concretely, the state space is $\mathcal{S}$, the action space is $\mathcal{A}$, the transition kernel $P(s' \mid s, a)$ gives the probability of moving to state s' after taking action a in state s, the reward function is $r(s, a) \in \mathbb{R}$, and the discount factor is $\beta \in (0,1)$. A policy $\pi(a \mid s)$ maps states to action distributions. The return is $G_t = \sum_{k\geq 0} \beta^k r_{t+k}$, the value is $V^\pi(s) = \mathbb{E}^\pi[G_t \mid s_t = s]$, and the Q-function is $Q^\pi(s,a) = \mathbb{E}^\pi[G_t \mid s_t = s, a_t = a]$. The optimal value satisfies the Bellman equation $V^\star(s) = \max_a\{r(s,a) + \beta\,\mathbb{E}[V^\star(s') \mid s,a]\}$. Reinforcement learning finds an approximately optimal policy from interaction with the environment rather than from a fully specified model. We refer to RL algorithms (value iteration, Q-learning, policy gradient, actor-critic, soft actor-critic, proximal policy optimisation, deep deterministic policy gradient) by name without re-deriving them; the algorithmic taxonomy is covered in Sections **??** and **??** of the survey.

0.1.2 Generalisations used in this chapter

Four generalisations of the MDP appear in the macro literature surveyed below.

A partially observed MDP (POMDP) supplements the MDP with an observation space $\mathcal{O}$ and an emission distribution $U(o \mid s)$. The agent does not see the state s directly; it conditions its policy on the observation history $o_{0:t}$, typically through a recurrent network or a sufficient statistic (a function of the history that preserves all decision-relevant information). POMDPs arise in macro whenever a household sees only prices or summary statistics rather than the full cross-sectional distribution, that is, the population-wide distribution of individual states such as wealth and productivity. The cross-sectional distribution matters to the household because equilibrium prices, and therefore its own budget constraint, depend on what the rest of the population is doing.

A Markov game extends the MDP to n agents who share the state but choose actions independently. Each agent $i \in \{1,\ldots,n\}$ has its own action space $\mathcal{A}_i$ and reward $r_i(s,a)$, where $a = (a_1,\ldots,a_n)$ is the joint action. Each agent treats the other agents' policies as part of a non-stationary environment and maximises its own return. Two-level games (a planner and a population of agents) are a special case.

A mean field game with common noise takes the continuum limit $n \to \infty$. A representative agent interacts with the cross-sectional distribution $\mu \in \Delta(\mathcal{S})$ of individual states and with a common noise $z \in \mathcal{Z}$ that captures aggregate shocks. The individual transition is $T(s' \mid s,a,\mu,z)$; the mean field evolves under $\mu_{t+1} = \Phi^\pi(\mu_t, z_t)$ for an operator Φ^π induced by the policy. The continuum limit makes the otherwise exponential per-agent state space tractable at the cost of losing finite-population effects.

A multi-objective MDP replaces the scalar reward r with a vector $r \in \mathbb{R}^d$. The solution concept is a Pareto front rather than a single optimum.

0.1.3 Equilibrium concepts

A policy $\pi_i^\star$ is a best response to the other agents' policies π_{-i} when $V_i^{(\pi_i^\star, \pi_{-i})}(s) \geq V_i^{(\pi_i', \pi_{-i})}(s)$ for every alternative π_i'. A Nash equilibrium is a profile $(\pi_1^\star, \ldots, \pi_n^\star)$ in which every component is a best response to the others. A mean-field Nash equilibrium is the continuum analogue, in which the representative agent's policy is a best response to the cross-sectional distribution it induces. A restricted perceptions equilibrium is a best response within a restricted policy class, for example policies that condition on prices but not on the full distribution; it is weaker than rational expectations and is the equilibrium concept SRL targets in Section 0.3. An ε-equilibrium weakens Nash by an ε slack on each player's incentive. Exploitability, $\sup_{\pi'} V^{\pi'} - V^{\pi}$ evaluated at the current mean field (the gap between the best response and the current policy), is the convergence metric used in MFG-RL; it upper-bounds the gap to a mean-field Nash equilibrium.

0.1.4 Notation summary

Throughout the chapter we use s for state, a for action, r for reward, π for policy, V and Q for values, β for the discount factor, P or T for the transition kernel, $\mu \in \Delta(\mathcal{S})$ for the cross-sectional distribution, $z \in \mathcal{Z}$ for the aggregate shock or common noise, p for the equilibrium price vector when explicit, θ for policy parameters, ϕ for auxiliary parameters when a second network is needed (a critic in single-agent RL, a planner in two-level RL), n for the number of agents in a Markov game, and i for an agent index. Algorithm-specific scalars (training-step counts N, T, trajectory counts E) are introduced inside the relevant algorithm boxes.

The chapter applies RL to macro models under two interpretations, introduced above and used without further comment hereafter. Under the solver interpretation, the policy network is a numerical device for finding an equilibrium that closed-form or DP-based methods cannot reach. Under the behavioural interpretation, the policy network is the household's actual cognition, learning a policy from realised utility under no prior model knowledge. Section 0.7 leans on the behavioural interpretation; Sections 0.2, 0.3, and 0.5 lean on the solver interpretation; the policy-design section uses one or the other depending on whether the planner is the focus.

A growing parallel literature uses deep learning rather than reinforcement learning to solve heterogeneous-agent and DSGE models by training networks to satisfy equilibrium equations as nonlinear regression objectives (Maliar et al., 2021; Azinovic et al., 2022; Azinovic-Yang and Žemlička, 2026; Kase et al., 2025a,b; Hall-Hoffarth, 2023; Carmona and Laurière, 2023; Fang et al., 2026; Thöni et al., 2025); Fernández-Villaverde et al. (2024) survey this line. The distinction is that those methods compute equation residuals offline, while RL trains the network from a reward signal under the agent's own policy. The chapter restricts attention to the RL lens and surfaces the equation-residual literature only as a comparison point where relevant.

0.2 RL for single-agent macro models

The simplest macroeconomic environment in which RL can be benchmarked against dynamic programming is the representative-agent stochastic real business cycle. The state space is two-dimensional, the analytical and numerical benchmarks are well understood, and the training dynamics expose how RL behaves on a problem for which DP remains the right tool. The interest is in the comparison rather than in any new claim about the RBC.

Atashbar and Shi (2023) train a representative household with deep deterministic policy gradient on a canonical stochastic RBC with log utility, Cobb-Douglas production, and AR(1) total factor productivity. The state is (K, A) with K capital and A productivity, the action is consumption C, and the transition is the capital accumulation equation together with the AR(1) for productivity. In both a deterministic variant without TFP shocks and a stochastic variant with shocks, the trained agent's consumption and investment policy lies close to the deterministic steady-state benchmark. The paper also documents training runs that fail to converge under high discount factors and suboptimal critic hyperparameters; the instability foreshadows the heterogeneous-agent setting in Section 0.3, where long-horizon credit assignment across an idiosyncratic-shock distribution amplifies the same problem. Section 0.4 returns to the same RBC with a side-by-side comparison across PPO, DDPG, value-function iteration, and a Blanchard-Kahn log-linearisation.

Shi (2022) trains an actor-critic agent on a stochastic optimal-growth environment with no a priori knowledge of preferences or the productivity process. The agent recovers the rational-expectations optimal saving rule asymptotically. Higher exploration accelerates learning at the cost of lower realised welfare during the exploration phase, which gives the explore-vs-exploit trade-off a macroeconomic interpretation that Section 0.7 returns to under the behavioural lens.

0.3 RL as a global solver for heterogeneous-agent macro models

Heterogeneous-agent (HA) macro models replace the representative agent of Section 0.4 with a continuum of households facing uninsurable idiosyncratic shocks on top of an aggregate shock. Canonical examples are Aiyagari (1994), Krusell and Smith (1998), and Kaplan et al. (2018). The state of the agent's problem now includes the cross-sectional distribution $\mu_t \in \Delta(\mathcal{S})$ of individual states, because equilibrium prices depend on μ_t and the distribution itself depends on the policy. Dynamic programming runs on an infinite-dimensional state and the value function lives on $\mathcal{S} \times \Delta(\mathcal{S}) \times \mathcal{Z}$. The traditional macro response summarises μ_t by a handful of moments (Krusell and Smith, 1998) or linearises around a stationary equilibrium (Reiter, 2009; Auclert et al., 2021). Both reductions impose ex ante which features of the distribution matter.

Yang et al. (2025) propose Structural Reinforcement Learning (SRL), which sidesteps the distribution entirely. The structural assumption is that the household knows its own budget constraint, preferences, and idiosyncratic shock process exactly. The only unknown part of the environment from the household's point of view is the joint process for equilibrium prices and aggregate shocks. SRL trains the policy by simulating the full economy and treating prices as exogenous to the agent through a stop-gradient. The sequence-space view of Auclert et al. (2021) justifies replacing μ in the agent state with the equilibrium price sequence: under rational expectations the price path is itself a sufficient statistic for the part of μ that affects forward-looking household decisions, because the household's budget constraint depends on μ only through equilibrium prices. Han et al. (2022), the immediate predecessor of SRL, kept the full cross-sectional distribution μ in the agent state via a deep-network policy that conditions on μ directly; SRL replaces μ with prices.

The SRL household's state is $s_t = (b_t, y_t, z_t, p_t)$, with b_t wealth, y_t the idiosyncratic productivity shock, z_t the aggregate shock, and p_t the equilibrium price vector. The action is $a_t = c_t$, consumption, with savings b_{t+1} determined residually by the budget. The next state is generated by three structural pieces: the budget constraint for b_{t+1}, the exogenous Markov chain for (y_{t+1}, z_{t+1}), and the market-clearing update $p_{t+1} = P^\star(\mu_{t+1}, z_{t+1})$ that the agent treats as exogenous. The reward is $r_t = u(c_t)$. The policy class is restricted to $\pi(s, z, p)$, which means the agent never conditions on μ at any step. Yang et al. (2025) call the resulting fixed point a sequential restricted perceptions equilibrium: given the price process induced by $P^\star$, the policy $\pi^\star$ maximises expected discounted utility within the restricted class, and market clearing holds at $p_t = P^\star(\mu_t, z_t)$ for every t. Agents' price beliefs are statistically consistent with realised prices on the equilibrium path, and the equilibrium is a fixed point in policy space rather than in the perceived-law-of-motion regression of Krusell-Smith.

The policy is parameterised as a tabular vector $\theta \in \mathbb{R}^{J\times K\times L}$ on the discretised (s, z, p) grid, and training uses the structural policy gradient algorithm of Algorithm 1. The cross-sectional distribution rolls forward to compute prices while a separate value-computation distribution rolls forward to score the policy. The Monte Carlo value is differentiated analytically by backpropagating through the known one-step transition matrix $A_\pi(z, p)$, the grid-restricted version of the transition kernel. Convergence is declared when the ℓ_∞ distance between successive policy iterates falls below a tolerance ε.

Yang et al. (2025) report runtimes on a single A100 GPU. Krusell-Smith converges in roughly 55 seconds and the authors report that its dynamics match the deep-learning rational-expectations benchmark. The Huggett model with aggregate risk, which has a nontrivial market-clearing condition for bonds, converges in roughly 75 seconds. A one-account HANK model with a forward-looking Phillips curve and sticky prices converges in roughly three minutes. Allowing agents to condition on a short price history rather than only the current price does not change the solution materially, which suggests that current prices encode most of the information relevant for behaviour in these calibrations.

0.4 Simulation: representative-agent RBC under DP and DRL

We check that RL matches dynamic programming on a problem for which DP is the right tool. The environment is a standard representative-agent stochastic RBC. The household chooses consumption C_t and saves the residual, with capital evolving as $K_{t+1} =$

Algorithm 1 Structural Policy Gradient (SPG) for HA macro models

Require: initial tabular policy $\theta_0 \in \mathbb{R}^{J \times K \times L}$ on the (s, z, p) grid; step sizes $\{\eta_k\}$; trajectory count N; horizon T; initial distributions ψ_g, ψ_z; convergence tolerance ε

for $k = 0, 1, 2, \dots$ **do**
  Sample N initial conditions $(\mu_0^n, z_0^n) \sim (\psi_g, \psi_z)$ for $n = 1, \dots, N$
  Initialise the value-computation distribution $d_0^n = d_0$, uniform over the individual-state grid
  for $n = 1, \dots, N$ **do**
    for $t = 0, 1, \dots, T-1$ **do**
      Compute the equilibrium price $p_t^n = P^*(\mu_t^n, z_t^n)$ from market clearing, with a stop-gradient applied to p_t^n
      Roll the cross-sectional distribution: $\mu_{t+1}^n = A_{\pi_\theta}(z_t^n, p_t^n)^\top \mu_t^n$ (used only to update prices)
      Roll the value-computation distribution: $d_{t+1}^n = A_{\pi_\theta}(z_t^n, p_t^n)^\top d_t^n$ (used only for the value)
      Sample the next aggregate shock $z_{t+1}^n \sim \Xi(\cdot \mid z_t^n)$
    end for
  end for
  Form the Monte Carlo value $\widehat{v}^\pi(\theta_k) = \dfrac{1}{N} \sum_{n=1}^{N} \sum_{t=0}^{T-1} \beta^t \left\langle d_t^n,\, u^{\pi_\theta}(z_t^n, p_t^n) \right\rangle$
  Compute the analytic gradient $\nabla_\theta \widehat{v}^\pi(\theta_k)$ by backpropagating through A_{π_θ} with prices held fixed by the stop-gradient
  Update $\theta_{k+1} \leftarrow \theta_k + \eta_k \nabla_\theta \widehat{v}^\pi(\theta_k)$
  if $\|\theta_{k+1} - \theta_k\|_\infty < \varepsilon$ **then**
    return θ_{k+1}
  end if
end for

$(1-\delta)K_t + A_t K_t^\alpha - C_t$ and total factor productivity following $\log A_{t+1} = \rho \log A_t + \varepsilon_{t+1}$ for $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2)$. Utility is $u(C_t) = \log C_t$. Parameters are calibrated as $\beta = 0.96$, $\alpha = 0.36$, $\delta = 0.10$, $\rho = 0.95$, $\sigma = 0.007$, with horizon $T = 200$. The deterministic steady state is $K^\star = 4.29$, $C^\star = 1.26$.

The state is $s = (K, A)$, the action is $a = C$, the reward is $r = \log C$, and the transition is the capital accumulation equation together with the AR(1) for TFP. Four methods are compared: a Blanchard-Kahn log-linearisation around the deterministic steady state (KPR), value-function iteration on a 400×41 tensor grid with a Tauchen-discretised TFP process (VFI), proximal policy optimisation with a Gaussian actor (PPO), and deep deterministic policy gradient (DDPG).[1]

Table 1 reports mean episode return, standard error, and policy mean-squared error

[1] Both RL methods use a 64-64 multilayer perceptron for actor and critic. Each is trained from scratch with 10 independent seeds. Every method is evaluated on a fixed set of 30 initial conditions $(K_0, A_0) \sim \mathcal{U}[0.5, 8] \times \mathcal{U}[0.95, 1.05]$ and shock paths $\{\varepsilon_t\}_{t=1}^T$, identical across methods, so cross-method differences are not driven by evaluation noise. The training budgets and convergence-monitoring schedules are recorded in the simulation script.

against VFI. KPR matches VFI to within 0.001 in policy MSE and within statistical noise on mean return. PPO converges to within the standard error of VFI with a policy MSE of 0.009. DDPG attains a mean return within 2% of VFI but with roughly double the cross-seed standard error and a policy MSE of 0.037, consistent with the known sensitivity of deterministic policy gradient methods to hyperparameter and initialisation choices. Figure 1 traces the learning curves of PPO and DDPG against the VFI and KPR horizontal benchmarks. Both RL methods approach the analytical benchmark within their respective training budgets. The result is what one would hope on a model this small. It is not a demonstration that RL replaces dynamic programming; it is a check that RL does not fail it on the textbook case.

Method	Mean return	SE	Policy MSE vs VFI	Wall clock
KPR	45.88	0.587	0.0004	–
VFI	45.86	0.589	0.0000	13.5 s
PPO	45.82	0.843	0.0087	–
DDPG	45.17	1.476	0.0365	–

Table 1: Representative-agent RBC under four methods. Mean episode return averaged over 30 evaluation episodes with shared initial conditions and shock paths. Standard error across 10 training seeds for PPO and DDPG, across 30 evaluation episodes for KPR and VFI. Policy mean-squared error reported against VFI consumption on 1000 random states drawn uniformly from the evaluation support.

0.5 Mean Field Games as a large-population RL problem

The mean-field game generalisation of Section 0.1 applies when large populations of identical, anonymous agents interact through aggregate quantities. Direct multi-agent RL does not scale, since the joint state and action spaces grow exponentially in n. Taking the continuum limit reduces the problem to a single representative agent whose environment includes the population distribution $\mu \in \Delta(\mathcal{S})$ and the common noise z. The RL problem is to find a policy that is a best response to the distribution it induces under the noise process. Convergence to a mean-field Nash equilibrium is monitored by exploitability.

The Krusell-Smith environment appears here under a different lens than in Section 0.3. SRL conditions agents on the current price vector and treats it as a Markovian sufficient statistic; RSPG instead lets agents carry a recurrent memory of past observations, relaxing the Markov assumption and replacing restricted perceptions with a partially observ-

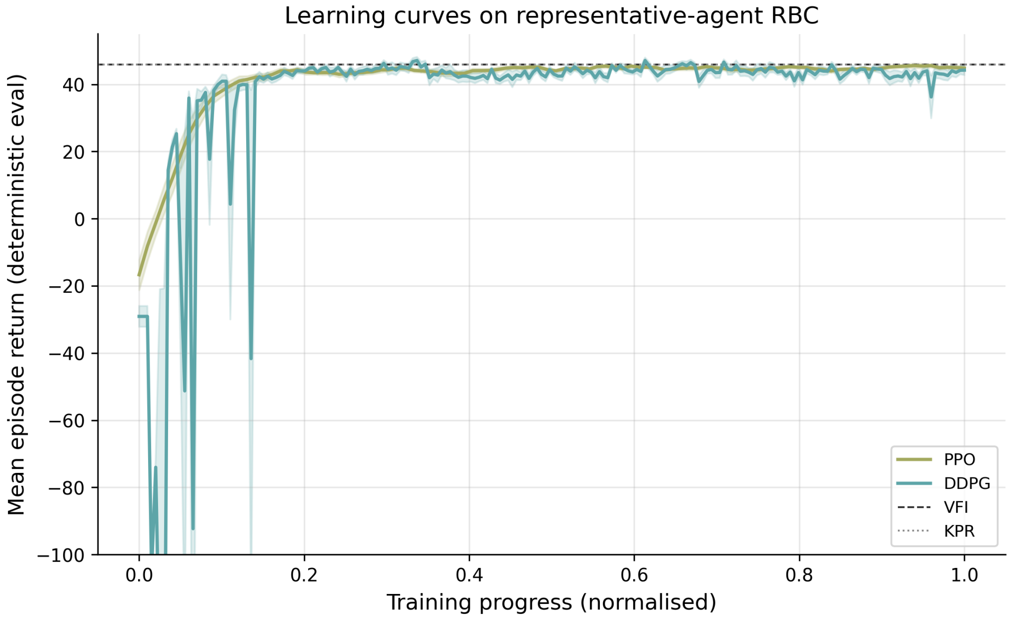

Figure 1: Mean deterministic-evaluation return as a function of normalised training progress for PPO (olive) and DDPG (cyan), shaded by cross-seed standard error across ten training seeds. Horizontal lines mark the VFI benchmark mean return (dashed black) and the KPR log-linear benchmark (dotted gray).

able mean-field Nash equilibrium. The two are alternative solution concepts for the same underlying household problem, not competing claims about the same equilibrium.

0.5.1 Recurrent Structural Policy Gradient

Wibault et al. (2026) propose the Recurrent Structural Policy Gradient (RSPG) algorithm for partially observable mean field games with common noise. In macroeconomic applications agents typically observe equilibrium prices but not the cross-sectional distribution, so the natural formulation is a POMFG. RSPG exploits the fact that individual transitions $T(s' \mid s, a, \mu, z)$ are known to the agent (the household knows its own budget constraint and idiosyncratic shock process), while the aggregate process for (μ, z) must be learned. The setting therefore sits between dynamic programming, which would integrate over both, and model-free RL, which would sample both.

In chapter notation the POMFG with common noise is a tuple $(p_{\mu_0}, p_{z_0}, \mathcal{S}, \mathcal{Z}, \mathcal{A}, T, \Xi, R, \beta, U, \mathcal{O})$, collecting initial distributions for the individual state and common noise, individual and aggregate state and action spaces, individual and common-noise transitions, reward, discount, and observation map. It combines the MFG primitives of Section 0.1 with the POMDP observation pair $(U, \mathcal{O})$ and a common-noise transition $\Xi(\cdot \mid z)$.

Algorithm 2 Recurrent Structural Policy Gradient (RSPG)

Require: initial recurrent policy θ_0; trajectories E; horizon T

for $k = 0, 1, 2, \ldots$ until convergence **do**
    Sample E common-noise trajectories $z^e_{0:T}$ with $z^e_0 \sim p_{z_0}$ and $z^e_{t+1} \sim \Xi(\cdot \mid z^e_t)$
    Roll forward $\mu^e_{t+1} = \Phi^{\pi_\theta}(\mu^e_t, z^e_t)$ analytically, with policy memory $h^e_{t+1} = \mathrm{RNN}(h^e_t, o^e_t)$
    Form the sample return $\widehat{J}(\theta_k)$ from (1) averaged over $e = 1, \ldots, E$
    Update $\theta_{k+1} \leftarrow \theta_k + \eta_k \nabla_\theta \widehat{J}(\theta_k)$ with gradients flowing through T and R only
end for
return θ^*

The mean field evolves under the analytic operator $\mu_{t+1} = \Phi^\pi(\mu_t, z_t)$ with (s, s') entry $\mathbb{E}_{a\sim\pi}[T(s' \mid s, a, \mu, z)]$. The RSPG architectural choice is to restrict the policy memory to a history of shared aggregate observations $o_{0:t}$, parameterised as the hidden state of a recurrent neural network. The reduced policy $\pi(a \mid s, h(o_{0:t}))$ retains history-dependence while keeping the asymptotic cost of the analytic mean-field update independent of the individual state.

For an initial common-noise distribution p_{z_0}, RSPG samples E trajectories in parallel, rolls out the endogenous mean field $\mu^\pi_{0:T}$ via the analytic operator Φ^π, and computes the discounted return

$$J^\pi \;=\; \mathbb{E}_{z_{0:T}\sim\Xi}\left[\sum_{t=0}^{T-1} \beta^t \left\langle \mu^\pi_t,\; R(\cdot,\, \pi,\, \mu^\pi_t,\, z_t)\right\rangle\right]. \tag{1}$$

Gradients flow through the individual transitions T and the expected-reward over actions. Gradients do not flow through the mean-field transitions; the analytic mean-field update is treated as a fixed environment, mirroring SRL's treatment of equilibrium prices.[2]

Wibault et al. (2026) evaluate RSPG on partially observable Linear-Quadratic, Beach-Bar, and Krusell-Smith environments. Across the three, RSPG attains the lowest or second-lowest exploitability among the algorithms compared and converges roughly an order of magnitude faster in wall-clock time than independent and recurrent PPO baselines. Finite-horizon agents under RSPG spend wealth just before episode end, pushing interest rates up and wages down. Memoryless variants of SPG and IPPO do not show this pattern. The paper positions RSPG as the first method to solve a partially observable Krusell-Smith model in which agents observe prices but not the cross-sectional

[2] The training loop is parameterised by the common-noise-trajectory batch size E, the rollout horizon T, and the RNN hidden dimension. Convergence is monitored by exploitability, the gap between the value of a best-response policy and the value of the agent's current policy at the current mean field. The exploitability metric requires a separate best-response oracle that Wibault et al. (2026) train as a single-agent RL agent against the frozen mean-field path; reported exploitabilities are therefore upper bounds.

distribution.

0.5.2 Hierarchical Stackelberg mean-field RL

Mi et al. (2025) extend the mean-field formulation of Section 0.5.1 from a symmetric population to a hierarchical setting in which a government (leader) sets policy first and a continuum of heterogeneous individual followers optimises against the announced policy. The framework formalises a Dynamic Stackelberg Mean Field Game (DSMFG) as the tuple $(\mathcal{S}^l, \mathcal{S}^f, \mathcal{A}^l, \mathcal{A}^f, P, r^l, r^f, T)$ collecting leader state space, follower state space, leader action space, follower action space, joint transition, leader reward, follower reward, and horizon. A mean field $L_t \in \Delta(\mathcal{S}^f \times \mathcal{A}^f)$ summarises the population state-action distribution.

The follower's best response to a leader policy π^l is the representative-agent policy

$$\pi^{f,*}(\pi^l) \;\in\; \arg\max_{\pi^f} \; \mathbb{E}\left[\sum_{t=0}^{T} \beta^t \, r^f(s_t^f, a_t^f, L_t, \pi_t^l)\right], \tag{2}$$

under the McKean-Vlasov fixed point that requires L_t to be the distribution induced by $\pi^{f,*}(\pi^l)$ at t. The leader then chooses

$$\pi^{l,*} \;\in\; \arg\max_{\pi^l} \; \mathbb{E}\left[\sum_{t=0}^{T} \beta^t \, r^l(s_t^l, a_t^l, L_t)\right] \tag{3}$$

anticipating the follower best response. The dynamic feedback is critical, as a static Stackelberg formulation cannot capture the temporal coupling that arises when followers observe and respond to the leader's policy over T periods.

The Stackelberg Mean Field Reinforcement Learning (SMFRL) algorithm trains both levels jointly under centralised training with decentralised execution. A Stackelberg mean-field Q-function $\widetilde{Q}^l(s^l, a^l, L)$ scores leader actions given the population state-action distribution; a follower-shared Q-network conditions on the same mean field. Best-response updates alternate: fix the leader, train followers to a ε-best-response and update L; then fix followers, optimise the leader. The mean-field approximation reduces the pair-wise complexity of agent-agent and agent-government interactions from $O(N^2)$ to $O(N)$.

Mi et al. (2025) report that DSMFG scales to 1,000 followers where the prior multi-agent benchmark in the same TaxAI environment (Mi et al., 2024) ran at one hundred agents. On the optimal-tax benchmark the learned leader policy achieves a four-fold per-capita GDP gain over the analytical Saez optimum (Saez, 2001) and a nineteen-fold improvement over the static 2022 US federal income tax schedule, both within the calibrated TaxAI environment. Ablations confirm that removing either the Stackelberg hierarchy or the mean-field approximation loses welfare or training stability. Exploitability decreases monotonically over training, indicating convergence to an approximate Stackelberg mean-field equilibrium. DSMFG bridges the symmetric mean-field methods of Section 0.5.1 and the single-planner setting of Section 0.8, in that the leader-follower structure makes the planner an explicit second decision-maker without dropping the continuum-population approximation that keeps the problem tractable.

0.5.3 Offline, asynchronous, and non-stationary MFG-RL

Three recent algorithms extend MFG-RL along orthogonal axes. Brunnbauer et al. (2024) learn an equilibrium policy from a static dataset of past trajectories without online interaction, by adapting Munchausen online mirror descent to offline data with an overestimation-mitigating regulariser; their experiments are on crowd-exploration and navigation, not macro. Yang (2026) drop the synchronous-move assumption by replacing the mean-action statistic $\bar{a}$ with the population distribution $\mu \in \Delta(\mathcal{O})$, derive a TMF Bellman equation, and prove an $O(1/\sqrt{N})$ finite-population exploitability bound that holds for any batch size of acting agents. Magnino et al. (2025) consider non-stationary continuous-space MFGs and report roughly an order-of-magnitude reduction in sampling cost on their benchmarks via a fictitious-play loop with a time-conditional normalising flow on the equilibrium density.[3]

0.6 Simulation: a partially observable linear-quadratic mean field game

To make the RSPG mechanism concrete, we use a compact version of the partially observable Linear-Quadratic environment in Wibault et al. (2026). The individual state is $s \in \{0, \dots, 98\}$ in the public MFAX implementation, the action is $a \in \{-3, \dots, 3\}$, the common noise is $z \in \{-1, 1\}$, and the horizon is $T = 30$. The common noise is persistent within an episode. It shifts the whole population down early in the episode, is neutral in

[3] Peng and Wang (2025) apply PPO with generalised advantage estimation and target networks to mean-field control games, a hybrid that combines internal coordination as in mean-field control with external competition as in MFGs.

the middle, and shifts it in the opposite direction late in the episode. Let $\bar{s}_t = \sum_{s'} \mu_t(s')s'$ denote the population mean. The agent observes its own state and $o_t = \bar{s}_t$, but not the full distribution, the common-noise realisation, or the calendar time. The mean field evolves by the analytic population transition operator $\mu_{t+1} = \Phi^\pi(\mu_t, z_t)$.

The step reward follows the paper's quadratic form,

$$R_t(s, a, \mu, z) = -c_a a^2 + qa(\bar{s}_{t+1} - s) - \frac{\kappa}{2}(\bar{s}_{t+1} - s)^2, \tag{4}$$

with terminal reward $R_T(s, \mu, z) = -c_{\text{term}}(\bar{s}_T - s)^2/2$. The MFAX defaults that we adopt unchanged are $c_a = 0.5$, $q = 0.1$, $\kappa = 0.5$, and $c_{\text{term}} = 1.0$, with idiosyncratic-noise scale $\sigma = 1.0$ and common-noise sensitivity $\rho = 0.5$. The horizon $T = 30$ is the MFAX truncation; the discount factor $\gamma = 0.99$ used during training is finite-horizon-compatible and matches the paper's configuration. The implementation follows MFAX in evaluating the reward after the population transition, using $\bar{s}_{t+1}$, and using the terminal reward on the final transition. The reward encourages agents to coordinate with the moving population while paying an action cost.

The reported results come from the public MFAX implementation of the paper's two structural policy-gradient algorithms on this environment.[4] SPG uses the analytic mean-field update and a memoryless categorical policy that conditions on current state and current mean observation. RSPG uses the same analytic update, but encodes the sequence of shared mean-state observations with a GRU; the recurrent hidden state is independent of the individual state, as in the paper. Both methods are evaluated with the analytic mean-field path induced by the final policy. Approximate exploitability is computed by backward induction against that path, so lower values mean a smaller gain from deviating to a best response. We do not include the paper's sampled recurrent PPO grid in the table, because that baseline is a much heavier finite-agent RL experiment rather than the structural mean-field update studied here.

Table 2 reports exploitability, expected return, and training time across ten training seeds after sweeping the official learning-rate grid. RSPG has lower exploitability than SPG, which is the partial-observation lesson of the paper's LQ example: history helps

[4]Specifically MFAX commit `9acc1eb` (`https://github.com/CWibault/mfax.git`). We apply only mechanical Python 3.11 dataclass `default_factory` fixes and a stdout addition that prints the per-iteration mean policy return; the diff is checked in at `patches/mfax/py311_compat.patch` with a README. None of the patched lines touches the SPG, RSPG, or environment update rules.

infer where the common-noise episode is headed.[5] Expected returns are reported for completeness but should not be read as a welfare ranking across methods, since each method induces a different mean-field path. Figure 2 plots exploitability over training and the final exploitability sensitivity to the learning rate.

Method	LR	Exploitability	Expected return	Train time (s)
SPG	10^{-2}	86.64 ± 16.29	-1021.3 ± 22.3	24.39 ± 0.06
RSPG	10^{-3}	60.37 ± 4.11	-1186.9 ± 10.1	28.79 ± 0.58

Table 2: Partially observable Linear-Quadratic mean field game. Approximate exploitability is computed by a best-response dynamic program on the analytic mean-field path induced by the final policy. Expected return and training time are averaged over 10 training seeds, with standard errors across seeds. Learning rates are selected from the official MFAX grid $\{10^{-4}, 10^{-3}, 10^{-2}\}$ by mean final exploitability for each method.

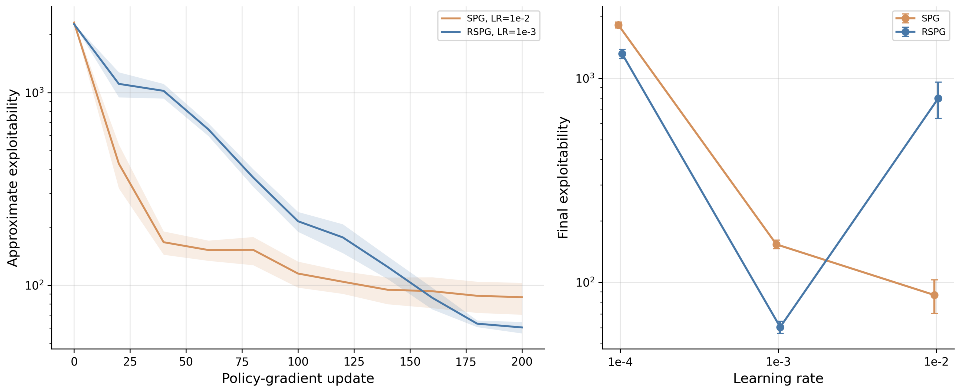

Figure 2: Official MFAX Linear-Quadratic grid. The left panel shows approximate exploitability over policy-gradient updates for the selected SPG and RSPG learning rates, shaded by cross-seed standard error across ten training seeds. The right panel shows final exploitability across the swept learning rates.

0.7 RL as a model of bounded rationality

The preceding sections used RL as a numerical device for finding equilibrium policies that closed-form or DP-based methods cannot reach. The interpretation switches here. The household itself learns from realised utility under no a priori knowledge of the income process, and the question is whether the resulting consumption behaviour matches empirical patterns that rational expectations does not.

[5]RSPG's mean final exploitability is non-monotone in the learning rate (1316, 60, 797 at $10^{-4}, 10^{-3}, 10^{-2}$), an interior optimum at 10^{-3}. SPG's minimum sits at the grid edge 10^{-2} (1824, 153, 87); a wider sweep might push SPG below its current 86.64 mean, but does not flip the ranking, since RSPG's regret floor at 60.37 is below SPG's interior best at this resolution.

Kaplowitz (2025) models a household solving a Markovian consumption-savings problem with cash-on-hand $x = y + Ra$, consumption share $\psi \in [0, 1]$, and savings residual $a' = (1 - \psi)x$, where $y \in \{y_e, y_u\}$ denotes employed and unemployed income drawn from a Bernoulli transition matrix $P_{yy'}$. The household does not know $P_{yy'}$ and learns an expected-continuation-value estimator $\widehat{EV}(y, a'; \phi)$ as a neural network with weights ϕ updated by stochastic gradient descent under a Robbins-Monro $1/\sqrt{t}$ learning-rate schedule. The Q-function is

$$Q(a, y, a') \;=\; u(\psi x) + \beta\, \widehat{EV}(y, a'; \phi), \tag{5}$$

and the consumption policy is a polynomial fit over Q. The agent is initialised at the rational-expectations solution to the same problem, so any deviation from rational behaviour reflects the learning dynamics rather than initial misspecification. The agent runs for fifty quarters with parameters drawn from the standard household-finance calibration (Krueger et al., 2016; Ganong et al., 2024 as cited in the source).

The same RL mechanism reproduces two empirical patterns that the combination of full-information rational expectations and borrowing constraints cannot generate jointly. First, the Ganong-et-al. marginal propensity to consume out of stimulus transfers is around 0.50 for previously-low-asset unemployed households and around 0.34 for previously-high-asset unemployed households, even when neither group is at a borrowing constraint. The simulated agents reproduce a gap of roughly the same magnitude (approximately 0.20). Second, the Malmendier-Shen scarring effect, in which past unemployment experience persistently lowers consumption controlling for current state, appears with the correct sign in the simulated paths. Both patterns trace to a single mechanism, value-function approximation errors that evolve with experience and generate ex-post heterogeneity from ex-ante identical agents. Belief-updating explanations, in which the household revises beliefs about $P_{yy'}$ after each unemployment spell, can produce the scarring effect but force consumption levels and MPCs to co-move in a way the data do not show.

Shi (2022) (Section 0.2) shows that a single-agent actor-critic on stochastic optimal growth converges to the rational-expectations optimal saving rule asymptotically; KAP's contribution is the non-asymptotic regime in which the agent has not yet converged. Kuriksha (2021) embeds heterogeneous neural-network learners in an Aiyagari-class economy. Each household updates its saving rule from its own realised consumption history, and the resulting wealth distribution is fat-tailed and exhibits the high average MPCs and excess sensitivity documented in the empirical literature, even without any single agent matching specific anomalies. The two papers together support the reading that RL households fail to satisfy rational expectations during learning in ways that reproduce stylised micro and macro facts, and that the mechanism is generic to value-function approximation rather than a peculiarity of any one architecture.

0.8 Multi-agent reinforcement learning for policy design

The first variant of RL for policy design treats the planner and the citizens as RL learners co-adapting in a shared environment. The planner's tax schedule, monetary rule, or climate-cooperation protocol is the action of one RL agent; the citizens' consumption, labour, or mitigation responses are the actions of other RL agents. Both sides learn from realised payoffs, so the planner sees the emergent strategic responses of the citizens to its policy choices. This setting is a Markov game in the sense of Section 0.1, with n heterogeneous agents whose actions are coordinated either by market clearing or by explicit search-and-match protocols. The contribution of RL is the discovery of mechanism designs and the surfacing of emergent behaviours that classical optimisation cannot reach. The single-planner variant in Section 0.9 drops the multi-agent training and treats the rest of the macroeconomy as a fixed environment.

0.8.1 MARL on canonical macro environments

Gabriele et al. (2026) build MARL-BC, an n-household RBC in which the representative-agent RBC and the Krusell-Smith mean-field economy appear as limit cases. Each household chooses a consumption fraction and labour supply at every period. Production is Cobb-Douglas in productivity-weighted aggregate capital and labour. In the chapter's Markov-game notation, the state is shared across agents and contains aggregate capital, labour, and the per-agent productivity draws; each agent's action is its consumption fraction and labour choice; each agent's reward is the log-plus-leisure utility

$$r_t^i \;=\; \log c_t^i + b\,\log(1-\ell_t^i). \tag{6}$$

The authors benchmark PPO, SAC, DDPG, and TD3 and report that the off-policy methods are more sample-efficient than PPO on the configurations they test.[6] With $n = 1$ the framework reproduces the textbook RBC solution. With a large population of ex-ante identical agents (in their largest run, on a 23×23 productivity grid yielding $n = 529$) it reproduces the Krusell-Smith mean-field wealth distribution and aggregate dynamics. With a population of heterogeneous agents it produces wealth and consumption distributions that mean-field methods cannot represent.

Curry et al. (2022) consider a richer micro-founded RBC with 100 worker-consumers, 10 firms, and a tax-setting government, formulated as a partially observable Markov game in which firms set prices and wages and goods are rationed proportionally if demand exceeds supply. Each agent type is trained by independent PPO with parameter sharing within type, accelerated by the WarpDrive GPU framework.[7] In the open-economy variant with an external export market, the model exhibits multiple qualitatively distinct ε-meta-equilibria.

0.8.2 RL agents inside macro ABMs

Brusatin et al. (2024) extend an existing macroeconomic ABM with capital and credit, in which 1,000 worker-households, 20 capital-goods firms, 100 consumption-goods firms, and one bank interact through search-and-match consumption markets, a labour market, and a credit market. The consumption-goods firms in the original ABM follow a hand-coded trend-following rule for setting prices and target quantities. The authors replace N of these 100 firms with tabular Q-learners.

Each RL firm i has state $s_{i,t}$ consisting of its log price-delta and log stock-delta relative to market averages, both binned on a finite grid; the action $a_{i,t}$ is a discrete multiplicative adjustment to next-period price and quantity; the reward $r_{i,t}$ is profit normalised by assets, with a bankruptcy penalty. The Q-table update is the standard

[6] Sample efficiency is reported in terms of environment steps to reach a fixed return threshold. The exact threshold and the number of seeds vary across the three population configurations, $n = 1$, $n = 20$ ex-ante identical, $n = 20$ heterogeneous. The paper does not provide a single hyperparameter table for the four algorithms.

[7] Without a curriculum, independent learners fail to recover non-trivial behaviour. A multi-phase curriculum that stages consumer, firm, and government training recovers non-trivial ε-meta-equilibria. The ε metric is evaluated by training single-agent best-response oracles against frozen others, so reported values are upper bounds.

tabular Q-learning rule

$$Q_{i,t+1}(s_{i,t}, a_{i,t}) \;=\; (1-\alpha)\, Q_{i,t}(s_{i,t}, a_{i,t}) + \alpha \left[r_{i,t} + \beta \max_{a'} Q_{i,t}(s_{i,t+1}, a') \right] \tag{7}$$

with learning rate α and ε-greedy exploration.[8]

Brusatin et al. (2024) report that RL firms learn three distinct profit-maximising strategies depending on competition level and the share of rational firms. Independent learners with separate Q-tables segregate into distinct strategic groups, raising aggregate market power and profits. Raising the share of RL firms always raises total output in the configurations tested, sometimes at the cost of higher output volatility.

0.8.3 Two-level deep RL for optimal taxation

Zheng et al. (2022) propose the AI Economist, a two-level deep RL framework for optimal taxation. The inner level is a population of agents who maximise discounted post-tax utility; the outer level is a social planner who sets a tax schedule to maximise a social-welfare function. Both levels are parameterised by deep neural networks and trained with proximal policy optimisation.

In the chapter's Markov-game notation, agent i has policy π_θ and solves

$$\max_{\pi_\theta} \; \mathbb{E} \sum_{t=0}^{H} \beta^t \, u_i(C_{i,t}, L_{i,t}), \tag{8}$$

where u_i is isoelastic in post-tax income $C_{i,t}$ with curvature $\eta > 0$ and linear in cumulative labour $L_{i,t}$; post-tax income depends on a bracketed tax schedule $\mathcal{T}(z;\tau)$. The planner has policy π_ϕ and solves

$$\max_{\pi_\phi} \; \mathbb{E} \sum_{t=0}^{H} \beta^t \, \mathrm{swf}_t, \tag{9}$$

with swf_t either inverse-income-weighted utilitarian welfare or the product of equality and productivity.

The two-level formulation is unstable because each side's effective MDP is non-stationary in the other side's policy. Zheng et al. (2022) stabilise training with two mechanisms.

[8] The grid discretisation, the exploration-rate schedule, and the training-step budget per firm are left as configurable inputs in the published code. The shared-policy variant uses a single Q-table across all RL firms; the independent-policy variant maintains a separate Q-table per firm. The market-competition level is controlled by the number of firms a consumer compares prices across.

Algorithm 3 Two-level Reinforcement Learning, after Zheng et al. (2022)

Require: initial agent parameters θ, planner parameters ϕ; sampling horizon H; tax-period length T; inner learner $\mathcal{A}$ (PPO in this work)

Initialise curriculum and planner-entropy schedules
while not converged **do**
  Reset world state s_0 and replay buffers $\mathcal{D}$ (agents), $\mathcal{D}_p$ (planner)
  for $t = 0, \ldots, H$ **do**
    if $t \bmod T = 0$ **then**
      Planner samples tax rates $\tau \sim \pi_\phi(\cdot \mid o_{p,t}, h_{p,t})$
    end if
    Each agent samples action $a_{i,t} \sim \pi_\theta(\cdot \mid o_{i,t}, h_{i,t})$ in parallel
    Step the environment; compute post-tax rewards; deliver the planner reward at tax-period boundaries
    Append transitions to $\mathcal{D}$ and $\mathcal{D}_p$
  end for
  Update θ via $\mathcal{A}$ on $\mathcal{D}$; update ϕ via $\mathcal{A}$ on $\mathcal{D}_p$
  Advance curriculum and entropy schedules
end while
return (θ, ϕ)

A curriculum gradually introduces labour costs and then taxes, so that early in training the agents experience low costs and explore freely. Entropy regularisation on π_ϕ keeps the tax schedule diffuse, so that agents experience a wide range of tax rates and learn to condition on them. These mechanisms stage training in two phases: agents adapt to random taxes first; the planner then enters and the two levels co-adapt.[9]

Zheng et al. (2022) test the framework in two environments. In a one-step economy with closed-form Saez optimum (Saez, 2001), the AI Economist's learned tax schedule matches the Saez optimum to within statistical noise. In the four-agent Open-Quadrant variant of the Gather-Trade-Build spatial environment, the learned schedule reportedly outperforms a Saez baseline calibrated to the same environment by 8% on utilitarian welfare and 12% on the product of equality and productivity. The schedule is non-monotonic, with highest marginal rates on middle income brackets. Emergent behaviour includes specialisation by skill into gatherers and builders and intertemporal tax avoidance by high-skill agents. The result is suggestive rather than decisive: the two-level RL planner discovers tax schedules that the analytical Saez framework does not, but the wel-

[9] Both networks are LSTM-based and trained with PPO; the agent and planner have separate replay buffers $\mathcal{D}$ and $\mathcal{D}_p$. The planner acts every T environment steps at a tax-period boundary; agents act every step. Curriculum and entropy schedules are advanced between training iterations and reported in the paper's experimental appendix. The reported wall-clock cost is several thousand GPU-hours for the Gather-Trade-Build runs, the largest training budget of the anchor papers in this chapter.

fare ranking depends on the choice of SWF and on the absence of behavioural responses outside the modelled action space.[10]

0.8.4 Scaling MARL-based optimal taxation

Mi et al. (2024) extend the AI Economist agenda to a Bewley-Aiyagari environment calibrated to US data, scaling from the four-to-ten-agent Gather-Trade-Build sandbox of Zheng et al. (2022) to ten thousand heterogeneous households. Households face idiosyncratic labour-income shocks and choose consumption, saving, and labour supply; firms produce with Cobb-Douglas technology; a financial intermediary intermediates household savings to firm investment; the government sets income-tax brackets and lump-sum transfers. The framework benchmarks seven multi-agent RL algorithms (DDQN, MAAC, MADDPG, BiCNet, and centralised-training/decentralised-execution variants) against a genetic-algorithm baseline and a dynamic-programming single-planner baseline. The seven MARL methods uniformly outperform the classical baselines on welfare metrics in the configurations tested. Households trained under MARL exhibit emergent tax-avoidance strategies, a strategic-response phenomenon that the smaller-scale spatial environment of Zheng et al. (2022) cannot surface. The finding is a scale-up rather than a methodological alternative; AIE remains the conceptual anchor for two-level RL policy design.

0.8.5 Multi-region RL on RICE-N

Climate-economy policy combines multiple regions interacting through global temperature and trade, no central authority enforcing cooperation, and a multi-decade horizon. The benchmark family of models is the integrated assessment model, which couples a climate block to an economic block and informs IPCC reports. The canonical IAMs are DICE (Nordhaus, 1992) and its regional extension RICE. RICE-N replaces the optimal-control solver inside RICE with a multi-agent system in which each region is a strategic learner.

In chapter notation RICE-N is a Markov game with n regions. The state $s_t = (s_t^{\text{nat}}, s_t^{\text{soc}})$ has a natural component, the atmospheric carbon mass and global mean tem-

[10] Tacchetti et al. (2025) survey the deep mechanism design landscape, recommending bilevel optimisation as the central stabilisation strategy and noting that human-response models built from imitation data are not robust to strategic adversaries who anticipate being modelled.

perature, and a social component collecting each region's capital, technology, population, trade balance, and the negotiation state. Region i's action $a_{i,t}$ is a tuple of a savings rate, mitigation rate, import-demand and import-tariff vectors, and a vector of negotiation actions. The transition factorises into a natural block (carbon mass and temperature evolve by the carbon-cycle and energy-balance equations inherited from DICE, with aggregate emissions $\sum_i E_{i,t}$ as the only social-to-natural channel) and a social block (Cobb-Douglas output scaled by a damage factor $1 - \Omega(T_t)$, with capital accumulating in the standard way and the negotiation state updating under the protocol in force). Region i's reward is the isoelastic utility of per-capita consumption, where consumption is an Armington aggregate of domestic and imported goods net of foreign tariffs.

RICE-N uses RL rather than DP because the game has no central planner, is non-stationary from any one region's perspective, has a combinatorial negotiation action space, and is only partially observed; classical RICE sidesteps all four by computing a single-planner Nash equilibrium. The agents themselves remain utility-maximisers; the RL is the method that finds their strategic policies, not a claim that regions are boundedly rational. The cost of the RL solution is the loss of the optimality certificate and the seed-independent reproducibility that the classical RICE optimum provides.

The contribution of RICE-N is to formalise the negotiation stage that precedes economic activity at each timestep, with one timestep representing five years. Zhang et al. (2025) compare two negotiation protocols. Bilateral Negotiation has each region propose a pair $(\hat{m}_i, \hat{m}_j)$ to every other region j, stating its own promised mitigation rate and the rate it requests; regions accept or reject; each commits to the maximum mitigation rate across accepted proposals. The Basic Club protocol (Nordhaus, 2015) proposes a club rate $\hat{m}_i$; regions accept or reject; a club forms with a common minimum rate, and non-members face a tariff proportional to the gap. Each region's policy is a neural network with weights shared across regions and region-specific inputs, trained with advantage actor-critic.[11]

Three findings carry policy interpretation that the classical DICE and RICE optimi-

[11]Both negotiation protocols reduce temperature growth at the cost of a small drop in production, while distributing emission-reduction costs more equitably than the no-negotiation baseline in the published runs. Zhang et al. (2025) are deliberately non-committal on what equilibrium concept the learners reach; the framework produces a family of outcomes indexed by protocol design rather than converging to a single solution. The AAAI workshop predecessor (Zhang et al., 2022) introduced the modelling framework without the Basic Club protocol; Heitzig et al. (2023) proposes a Conditional Commitments mechanism intended to enhance participation stability.

sations cannot produce. A climate club enforced by border tariffs is self-sustaining in the configurations tested. The negotiation institution is itself a policy lever, in that Bilateral Negotiation and Basic Club reach materially different climate and economic outcomes from the same underlying economy. Both protocols lower the cross-region inequality of abatement cost and mitigation rate relative to the no-negotiation baseline, and the authors caution that a uniform border tariff can still operate as a de facto carbon tax on developing regions absent redistribution or technology transfer.

0.8.6 Multi-objective MARL for equitable policy

Biswas et al. (2025) keep the multi-region structure of RICE-N but change the reward. JUSTICE incorporates the economy, damage, and abatement modules from the 57-region RICE50+ integrated assessment model and couples them with the FAIR climate emulator. It is a multi-objective multi-agent Markov decision process in which the regions share a team reward that is a two-component vector, the negative of global mean temperature and global net economic output, both to be maximised. Training uses a multi-objective multi-agent actor-critic that decomposes the vector-reward problem into scalarised single-objective problems via weighted-sum scalarisation.[12] JUSTICE and RICE-N together frame the question of whether RL converges to one equilibrium or many. RICE-N reaches one realisation at a time from a family of possible equilibria; JUSTICE enumerates the Pareto frontier directly. A single-objective RICE optimiser collapses the climate-equity trade-off into one weighted-sum welfare function, fixing the trade-off before the policymaker sees it; the Pareto-front output leaves it open.

0.9 Single-planner reinforcement learning for policy design

The second variant of RL for policy design treats the planner as the sole RL agent. The rest of the macroeconomic environment is a fixed DSGE, a fitted VAR, or an empirically calibrated structural model populated by rational households whose responses are taken as given. The contribution of RL is the ability to learn nonlinear and state-dependent reaction functions in environments where closed-form optimal control is intractable. The setting is closer to optimal control theory than to mechanism design, with RL in the role

[12] Biswas et al. (2025) sample one hundred weight vectors, train one policy per weight, and collect the non-dominated policies into the solution set. The output is a frontier of policies rather than a single policy. A Gini-based equity index across regions is reported alongside the climate-output frontier.

that nonlinear model predictive control plays in classical engineering applications.

0.9.1 Single-planner regime learning in a monetary DSGE

Chen et al. (2023) apply deep RL to the Benhabib et al. (2001) model of monetary-fiscal interaction. A representative household solves

$$\max\, \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(c_t) + v(m_t) - w(n_t) \Big] \quad \text{s.t.} \quad c_t + m_t + b_t = w_t n_t - \tau_t + \frac{R_{t-1} b_{t-1} + m_{t-1}}{\pi_t}, \tag{10}$$

where $m_t = M_t/P_t$ is real money, $b_t = B_t/P_t$ is real bonds, τ_t is a lump-sum tax, w_t is the real wage, $\pi_t = P_t/P_{t-1}$ is gross inflation, and output is linear in labour, $y_t = \varepsilon_t^y n_t$, with ε_t^y an i.i.d. technology shock. Fiscal policy follows a Leeper-style rule $\tau_t = \gamma_0 + \gamma\, b_{t-1} + \varepsilon_t^\tau$ and monetary policy follows a global nonlinear interest-rate rule $R_t = f(\pi_t)\, \varepsilon_t^R$ with f strictly positive and strictly increasing. The deterministic Fisher equation $R = \pi/\beta$ intersects $f(\pi)$ at the inflation target $\pi^\star$ and at a liquidity trap π_L.[13]

The household's state is $s_t = (m_{t-1}, b_{t-1}, \pi_{t-1}, c_{t-1}, n_{t-1}, \varepsilon_t^\tau, \varepsilon_t^R, \varepsilon_t^y)$, the action $a_t = (c_t, b_t, n_t)$ is a continuous tuple of consumption, bond holdings, and labour, the reward $r_t = u(c_t) + v(m_t) - w(n_t)$ is the instantaneous utility, and the transition $P(s_{t+1} \mid s_t, a_t)$ is induced by the household's intertemporal Euler equation

$$u'(c_t) \;=\; \beta\, R_t\, \mathbb{E}_t \left[\frac{u'(c_{t+1})}{\pi_{t+1}} \right] \tag{11}$$

together with the equilibrium fiscal-monetary rules and the exogenous shock processes. The policy is learned by soft actor-critic (Haarnoja et al., 2018).[14]

The substantive claim is that the set of monetary-fiscal regimes in which a household can converge to a stationary policy is strictly larger under deep RL than under the recursive-least-squares adaptive-learning benchmark of Evans and Honkapohja (2005).

[13] Each policy rule is locally active or passive depending on the sign of a coefficient. Fiscal policy is active when $\gamma < \beta^{-1} - 1$. Monetary policy is locally active at inflation π when $f'(\pi) > \beta^{-1}$. The cross-product of the two stances generates four regimes, of which the doubly-passive case is locally indeterminate under rational expectations and the doubly-active case is locally explosive. Evans and Honkapohja (2005) show that recursive-least-squares adaptive learning reaches only the determinate regimes.

[14] SAC is an off-policy maximum-entropy actor-critic. The Gaussian actor π_θ and the Q-function Q_ϕ are deep neural networks trained by stochastic gradient descent on minibatches drawn from a replay buffer. The critic update follows the temporal-difference loss with a target network and the actor update is the reparameterised maximum-entropy gradient. Chen et al. (2023) report that the trained policy converges to a stationary outcome in all four monetary-fiscal regimes, including the doubly-passive case (locally indeterminate under rational expectations) and the doubly-active case (locally explosive). The interpretation is that the deep policy is constrained by the global utility-maximisation problem rather than by the linearised dynamics around any one steady state.

RL extends learnability to the indeterminate and explosive regimes that adaptive learning cannot reach.

0.9.2 Benchmarking RL on monetary policy

Wang et al. (2025) compare nine RL algorithms on a monetary-policy MDP fitted to US Federal Reserve quarterly data (1955–2025). In chapter notation the state is $s_t = (\pi_t, U_t, \mathrm{gap}_t, R_t)$ comprising inflation, unemployment, the output gap, and the policy rate; the action a_t is a discrete adjustment to the policy rate; the reward encodes the Fed's dual mandate as a quadratic loss in deviations from a 2% inflation target and a 4.5% natural unemployment rate. The transition is a linear-Gaussian VAR fitted to historical transitions.[15] Standard tabular Q-learning achieves the best mean discounted return, outperforming both the enhanced RL methods and a Taylor-rule baseline. The dynamics are linear-Gaussian rather than a structural DSGE and the paper has not been peer-reviewed, so the finding is preliminary. It suggests, no more strongly than that, that sophisticated RL machinery does not dominate the combination of simple rules and tabular RL in low-dimensional monetary-policy MDPs.

0.9.3 Single-planner RL on data-grounded monetary models

Hinterlang and Taenzer (2024) train a central-bank RL agent on a neural-network surrogate of the US economy estimated from quarterly data 1987Q3–2007Q2. The pipeline first fits transition equations from data, in linear (SVAR) and nonlinear (feedforward ANN) variants, and then trains an RL reaction function inside the fitted environment with the zero lower bound, asymmetric central-bank preferences, and the dual mandate baked into the reward. The RL-optimised linear rule reduces the central-bank loss by more than thirty-five per cent relative to common Taylor-style rules and the realised federal funds rate over the sample window. The nonlinear ANN reaction function extends this to a forty-three per cent reduction. A cross-validation exercise feeds the learned rules into eleven alternative DSGE models from the Macroeconomic Model Database and finds smaller unconditional inflation and output variances under the RL rules than under the Taylor benchmarks, a generalisation result absent from the MPU comparison above.

[15] The nine algorithms span tabular Q-learning variants, SARSA, actor-critic, deep Q-networks, Bayesian Q-learning, and a POMDP particle-filter variant. The training and evaluation protocols are aligned across algorithms, with each agent trained on the same fitted VAR dynamics and evaluated on a held-out window.

Khundadze and Semmler (2025) extend the single-planner setup to the Eurozone, using soft actor-critic to control inflation, the output gap, and debt in a two-bloc (North-South) macro environment with asymmetric shocks. The RL policy is benchmarked against a nonlinear Model Predictive Control (NMPC) baseline. SAC achieves comparable or better stabilisation on average and is more robust under asymmetric shocks because the learned feedback policy generalises where NMPC must re-solve an open-loop trajectory each period. Cross-bloc transfers emerge as a learned component of the joint fiscal-monetary policy.

0.10 Discussion

0.10.1 Four roles, one toolbox

Reinforcement learning enters macroeconomics in four distinct roles covered above. As a global solver, structural policy gradient methods solve heterogeneous-agent models on timescales that classical projection and perturbation cannot reach. As a game-theoretic formalism, mean-field RL handles partial observability and hierarchical Stackelberg structure. As a model of the household, RL learning dynamics reproduce empirical anomalies that rational expectations with borrowing constraints cannot generate jointly. As a planner, RL discovers tax schedules, climate-cooperation mechanisms, and monetary-fiscal reaction functions that closed-form optimisation does not. The four roles share an algorithmic toolbox but answer different questions, and the empirical and theoretical literatures around each are at different stages of maturity. The current state is best read as early progress rather than as a settled methodology.

0.10.2 Common methodological issues

Reading across the lenses surfaces issues that no single section makes explicit. The first concerns convergence theory. Of the algorithms surveyed in this chapter, only the temporal mean-field construction of Yang (2026) carries a classical guarantee, with existence and uniqueness of the equilibrium under a discrete monotonicity condition, finite-population $O(1/\sqrt{N})$ Nash approximation regardless of the per-step batch size, and provable convergence of the policy-gradient iterate (the sequence of policy updates produced by the algorithm). The remaining methods, including Yang et al. (2025), Wibault et al. (2026), Kaplowitz (2025), Zheng et al. (2022), Chen et al. (2023), Mi et al. (2025), Han et al.

(2022), Brunnbauer et al. (2024), and Magnino et al. (2025), monitor convergence empirically by exploitability, L^∞ norms of policy updates, or Bellman residuals against application-specific tolerances. This is not a failure of rigour. The macro equilibrium concepts these algorithms target, including restricted perceptions equilibrium, mean-field Nash, Stackelberg mean-field, and co-adaptation (joint training of planner and agents against each other's current policy), live in spaces where classical fixed-point theorems (existence results such as Brouwer or Kakutani) either do not apply or do not yield computable bounds. The honest reading of the literature is that convergence here means training stopped changing the policy, not that the iterate provably reaches a model equilibrium.

The four lenses also target four different equilibrium concepts, and papers within a lens diverge further (Section 0.1). The restricted perceptions equilibrium of Yang et al. (2025) assumes agents condition on the current price vector as if it were Markov, which is not satisfied by true Krusell-Smith equilibrium prices but is what makes structural policy gradient tractable. Wibault et al. (2026) extends this to a partially observable mean-field Nash with a shared observation and recurrent memory. Mi et al. (2025) targets a Stackelberg mean-field with a single leader and a continuum of homogeneous followers. Zheng et al. (2022) trains planner and citizens by PPO against each other's current policy without naming the equilibrium the joint iterate is supposed to reach. Chen et al. (2023) accept whichever DSGE steady state the SAC iterate stabilises at, including the indeterminate and explosive regimes that adaptive learning cannot reach. Kaplowitz (2025) abandons equilibrium altogether in favour of single-agent Q-learning calibrated to micro-panel moments. A claim that RL solves Krusell-Smith therefore means different things under Yang et al. (2025) than under Han et al. (2022), where the same model is solved as a recursive equilibrium in which μ is summarised by a finite set of moments (moment closure), with neural-network value and policy functions.

Several recurring pitfalls follow from these design choices. Multiplicity is acknowledged but resolved silently by initialisation, through the warm-up phase of Yang et al. (2025), the two-phase entropy schedule of Zheng et al. (2022), and the regime-by-regime training of Chen et al. (2023); none of these papers tests initialisation robustness. Curriculum learning is empirically necessary but theoretically unmotivated, with Curry et al. (2022) showing independent multi-agent RL fails to recover non-trivial behaviour without one and Zheng et al. (2022) confirming the same pattern in two-level policy design. Hyperparameter sensitivity is severe and unevenly characterised, with Wibault et al. (2026) reporting that off-the-shelf IPPO and recurrent-IPPO require 81 to 243 configurations on partially observable Krusell-Smith while structural policy gradient methods converge with three; no paper offers transferable tuning guidance. Discount-factor sensitivity is essentially unexamined, with Atashbar and Shi (2023) flagging instability at high β but no paper sweeping β over the macro-realistic range. The single-agent to equilibrium gap is open, since the household learning rule of Kaplowitz (2025) reproduces marginal propensity to consume heterogeneity and scarring at the household level but has not been tested for consistency once embedded in an Aiyagari-style equilibrium where its policy would shift the cross-sectional distribution.

Three cross-cutting findings deserve to be surfaced separately from their parent sections. The first is that tabular and grid-based RL competes with or beats deep RL on macro problems. Wang et al. (2025) benchmarks nine algorithms on US Federal Reserve quarterly data and finds tabular Q-learning beats DQN, soft actor-critic, and Bayesian variants on a quadratic-loss monetary objective, and Yang et al. (2025) reports grid policies training faster than deep networks on Krusell-Smith. The ordering familiar from robotics and games does not transfer mechanically to macro, whose state spaces are low-dimensional once correctly compressed. The second finding is that RL reaches equilibrium regions adaptive learning cannot. Chen et al. (2023) shows deep RL stabilises in indeterminate and explosive DSGE regimes that least-squares learning rules out by E-stability (the expectational-stability condition under which a perceived law of motion is learnable by least-squares updating), suggesting the same toolbox extends the equilibrium set under bounded rationality rather than merely reproducing it. The third is sociological. The three methodological clusters covered in this chapter, structural solvers (Yang et al., 2025; Wibault et al., 2026; Han et al., 2022; Azinovic-Yang and Žemlička, 2026), behavioural single-agent learners (Kaplowitz, 2025; Kuriksha, 2021; Shi, 2022), and policy designers (Zheng et al., 2022; Mi et al., 2024; Chen et al., 2023; Hinterlang and Taenzer, 2024; Khundadze and Semmler, 2025; Wang et al., 2025), have evolved in parallel with limited cross-citation. The solver papers do not engage the behavioural ones, the behavioural paper of Kaplowitz (2025) cites no RL-macro work, and the two monetary policy papers (Hinterlang and Taenzer, 2024; Wang et al., 2025) appear unaware of each other.

The risk is that solver-driven equilibrium claims, behavioural calibration evidence, and policy-design exercises drift apart without methodological reconciliation.

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